QUANG-MINH DANG

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WORK EXPERIENCE

Techcombank – Private Commercial Bank Feb 2025 – Current
***Java Backend Engineer**:*

- Engineered core backend microservices of the **Retail Omni Channel (ROC)** system on AWS EKS, serving **10M+ retail customers** with high-availability and cloud-native architecture.
- Owned and delivered core features across strategic projects, including Family & Friend Banking (serving **1M+ monthly active users**), E-Ticket & Queue Management (digital transformation for **200+ branches**), Payment Agents (transforming **4,000+ WinMart/WinMart+ stores** into banking endpoints), Virtual Assistant Center (increased conversion rate by 103%) and Physical Gold Online Trading (aiming to be Vietnam market leader, launching Mar' 26)
- Integrated ROC system with complex internal systems (T24, Paylink, Notihub, MFT,...) and 3rd-party services (AWS Connect, Queuebee QMS, Equix GMS,...) to deliver end-to-end financial workflows.
- Designed and delivered enterprise-grade APIs, aiming for a **P95 latency of < 200ms**; maintaining **95%+ new code coverage** for new features.
- Ensured data correctness and auditability for all transaction-based flows by implementing strict idempotency, ACID principles, and race-condition prevention mechanisms.
- Architected high-performance backend services using **Redis/Caffeine** for hot-data caching and **Kafka** for resilient, asynchronous communication. Optimized persistence layers through JPA tuning and indexing, significantly improving overall system throughput.
- Utilized **Quartz** and **Kubernetes Jobs** to automate scheduled financial tasks, ensuring reliable execution and monitoring of background processes
- Leveraged comprehensive observability using **Dynatrace, OpenSearch, Grafana, AKHQ** for proactive system tracing and rapid incident response.
- Streamlined CI/CD pipelines and centralized configurations using **Jenkins**, **ArgoCD**, and **Spring Cloud Config** to ensure stable and frequent production releases.
- Adopted **AWS Kiro** - AI Coding Agent in the development process to accelerate debugging, refactor business logic, review code, significantly increasing development velocity and code reliability.

Vietnam People's Navy – Conscription Mar 2023 – Jan 2025
***Cryptologist** (11 mos):*

- Attached to the Navy Headquarters. Managed highly classified communications while maintaining high combat readiness, precise execution, and resilience under pressure.

***Squad Leader** (8 mos):*

- Supervised a 10-man squad, ensuring strict discipline, safety protocols, and effective teamwork during intense missions and field training.

Techcombank – Private Commercial Bank Oct 2022 – Mar 2023
Management Associate (TCB Future Gen - Technology stream)

- Outperformed over **1200 candidates** to be selected for a highly competitive **fast-track program**, gaining comprehensive exposure to enterprise-grade banking architecture and digital transformation strategies.
- Engineered automated cross-functional business workflows to enhance enterprise efficiency, reducing waiting time for the IT Operational Management team by 35%.
- Researched and developed Agile ability assessments for the IT Division, evaluated Key Result Areas across IT functions, and optimized SQL queries for management dashboards, reducing response time by nearly 30%.

AIV Group – Technology Startup Aug 2019 – Aug 2022
***Software Developer** (part-time)*

- Researched and deployed POC smart contracts on Rinkeby network for ContentBid (Web3 content marketplace), gaining highly relevant exposure to Web3 tooling and decentralized architectures.
- Optimized backend performance for the VnAlert (AI-powered news aggregator platform) by refining **Django ORM** query patterns. Improved system efficiency by reducing query overhead and leveraging PostgreSQL indexing through ORM-level optimizations.

- Developed AutoVid (automated video generation platform), building a robust Django backend, orchestrating a **12-node distributed render farm** using **Celery & RabbitMQ**, enabling the system to process and render **500+ high-quality videos daily** with automated resource scheduling.
- Tech stack: *Django (Python), PostgreSQL, Redis, Celery, RabbitMQ, MinIO, Docker, Solidity, Web3.js.*

InfoRe Technology – Technology Startup Incubator Jun 2018 – Aug 2019
Data Scientist Intern
- Developed a POC for an automated time attendance system for ABBank using **OpenCV**, **PyTorch**, and **YOLOv3** model, gaining foundational experience in identity verification mechanisms relevant to digital KYC flows.
- Processed large-scale visual datasets using NumPy and OpenCV, optimizing data pipelines for efficient model inference.
- Fostered continuous learning culture by conducting weekly AI seminars for developers and instructing an internal Python course for the Business & Data team.

Education

Hanoi University of Science and Technology (HUST) Sep 2017 – Sep 2022
- Major: Information and Communication Technology – Engineer's degree – GPA: 3.44/4.0
- Thesis: vToken system – generating and transferring tokens across multiple EVM blockchains

Tran Phu Gifted High School, Haiphong, VN Sep 2014 – Jun 2017

Skills

- **Backend & Frameworks:** Java (Spring Boot), Python (Django), JavaScript/TypeScript (NextJS).
- **Database & Caching:** PostgreSQL, MySQL, Caffeine, Redis, ElasticSearch, DynamoDB.
- **Architecture & API Design:** Microservices, Event-Driven Architecture, RESTful API Design, API-first development with OpenAPI, Clean Architecture, transaction handling (ACID), Idempotency, Reconciliation flows.
- **DevOps & Infrastructure:** Docker, Kubernetes, Git, CI/CD pipelines, Kafka, RabbitMQ, Proxmox.
- **Soft Skills:** High ownership, cross-functional collaboration, rapid adaptation & continuous learning, resilience under pressure.
- **Languages**: English (940/990 TOEIC), Vietnamese (Native).
- **Certificates:** AWS Certified Solutions Architect – Associate (SAA-C03)

Side Projects

TodoPro.xyz – AI-Agent friendly Personal Task Management SaaS Dec 2025
- Engineered a full-stack task management platform with AI capabilities, integrating third-party APIs for seamless workflow synchronization.
- Stack: *Django, Next.js, PostgreSQL, AWS S3, Python CLI, GitHub Actions (CI/CD), GitHub Copilot.*

SkillBoost.vn – AI Agent Skills Marketplace | Launching in Apr' 26 Feb 2026
- Stack: *Django, GCP Compute Engine, AI Coding Agents*

Awards & Activities

A-Class Academic Excellence Scholarship - HUST Jul 2022
- Achieved the highest-level academic scholarship for the school year 2021-2022

Top 10 – Developer Circles Vietnam Innovation Challenge Sep 2020
- Worked in a team of five to build a food recommendation mobile app

Top 5 – AI4VN National Hackathon Aug 2019
- Built a virtually try-on clothes web application powered by VITON model

Scholarship of FPT Young Talents - FPT Corporation Jan 2018
- Won scholarship and participated for a period of 2 years in the FYT center